04_DATABASE.md

TASK 04

DATABASE DESIGN & IMPLEMENTATION

Status

Mandatory

Owner

Database Architect

Supporting Guides

Project Director

Security Architect

Solution Architect

Backend Lead

Principal Engineer

Estimated Time

1 to 2 Days

Dependencies

00_PROJECT_CONTEXT.md

00_ENGINEERING_RULES.md

00_PROJECT_MEMORY.md

05_DATABASE_ARCHITECT.md

03_ARCHITECTURE.md

Mission

Design and implement a secure, scalable and production-ready relational database for the Secure Dance Academy Management System.

The database is the foundation of the application.

Every table should represent a real business entity.

Every relationship should enforce business rules.

Every constraint should protect data integrity.

Objectives

Design a normalized database.

Create a Prisma schema.

Generate database migrations.

Implement security policies.

Optimize performance.

Support future expansion.

Prepare the database for backend development.

Deliverables

Produce

Entity Relationship Diagram

Conceptual Data Model

Logical Data Model

Physical Data Model

Prisma Schema

Migration Files

Seed Data

Indexes

Constraints

Relationship Documentation

Data Dictionary

Backup Strategy

Restore Strategy

Database Security Report

Performance Review

Business Entities

Design complete models for

Users

Roles

Permissions

User Roles

Artists

Parents

Coach Assignments

Attendance

Attendance Sessions

Performances

Competitions

Performance Results

Injuries

Medical Records

Emergency Contacts

Activities

Notifications

Audit Logs

Refresh Tokens

Application Settings

Future expansion should require minimal schema changes.

Table Design

Each table must contain

UUID Primary Key

Created Date

Updated Date

Created By

Updated By

Status

Soft Delete Support where appropriate

Audit support

Meaningful naming

Avoid unnecessary columns.

Relationships

Create all relationships.

Examples

One User

↓

One Role

One Coach

↓

Many Artists

One Parent

↓

Many Children

One Artist

↓

Many Attendance Records

One Artist

↓

Many Performance Records

One Artist

↓

Many Injury Records

One Activity

↓

Many Participants

One User

↓

Many Audit Logs

Enforce relationships using foreign keys.

Normalization

Normalize to Third Normal Form.

Remove duplicate information.

Separate lookup data.

Avoid repeating groups.

Store reusable values once.

Document every normalization decision.

Constraints

Implement

Primary Keys

Foreign Keys

Unique Constraints

Check Constraints

Not Null Constraints

Business Constraints

Prevent invalid data at database level.

Indexing

Create indexes for

Email

Username

Role

Attendance Date

Coach

Artist

Performance Date

Status

Audit Timestamp

Search Fields

Review index effectiveness.

Avoid unnecessary indexes.

Prisma Schema

Generate a complete schema.

Use

Relations

Enums

Default Values

Indexes

Constraints

Cascade Rules

Keep schema readable.

Comment complex relationships.

Migrations

Generate versioned migrations.

Every migration must

Be reversible

Be documented

Be tested

Never edit existing migrations.

Create new migrations for every change.

Seed Data

Generate development seed data.

Administrator

Coach

Parent

Artists

Activities

Attendance Samples

Performance Samples

Notifications

Application Settings

Seed data should demonstrate system functionality.

Security

Protect the database.

Use

Prisma ORM

Parameterized Queries

Foreign Keys

Row Level Security

Least Privilege

Environment Variables

Secure Connections

Audit Logging

Never expose direct database access.

Row Level Security

When using Supabase

Enable RLS.

Design policies for

Administrator

Coach

Parent

Artist

Document every policy.

Apply deny-by-default.

Audit Logging

Create immutable audit records.

Log

User

Timestamp

Action

Entity

Old Value

New Value

IPAddress

Device

Audit records should never be editable.

Transactions

Use transactions for

Attendance

Medical Updates

Role Changes

Activity Registration

Profile Updates

Performance Results

Audit Logging

Ensure atomic operations.

Performance

Optimize

Indexes

Queries

Pagination

Filtering

Sorting

Reporting

Aggregation

Prevent N+1 queries.

Document optimization decisions.

Backup Strategy

Design

Daily Backup

Point-in-Time Recovery

Migration Recovery

Disaster Recovery

Backup Verification

Document restoration procedures.

Data Dictionary

Document every table.

Purpose

Columns

Data Types

Relationships

Constraints

Business Rules

Security Notes

No undocumented tables.

Database Testing

Verify

Relationships

Constraints

Indexes

Migrations

Seed Data

Transactions

RLS Policies

Query Performance

Integrity

Record testing results.

Documentation

Produce

ER Diagram

Relationship Diagram

Schema Documentation

Migration Guide

Seed Guide

Constraint Documentation

Index Documentation

Database Security Guide

Performance Report

Data Dictionary

Acceptance Criteria

The database should

Support every business requirement.

Protect data integrity.

Prevent duplicate records.

Scale efficiently.

Support reporting.

Support secure APIs.

Support audit logging.

Support future growth.

Update Project Memory

Mark

Database Design

Completed

Record

Schema Version

Migration Version

Entity Count

Relationship Count

Indexes

Security Policies

Known Limitations

Next Task

UI and UX Design

Database Review Checklist

Verify

Schema is normalized.

Relationships are correct.

Constraints are complete.

Indexes are optimized.

Prisma schema compiles.

Migrations succeed.

Seed data loads.

RLS policies work.

Audit tables exist.

Performance is acceptable.

Documentation is complete.

Definition of Done

The database is production ready.

All business entities exist.

Relationships enforce business rules.

Security policies are implemented.

Prisma schema is complete.

Migrations execute successfully.

Seed data is available.

Documentation is complete.

The database is approved for backend development and should not require structural redesign during later project phases.